COS326 Semester Test 1 2026

Final Grade

60 / 90

Your instructor added question feedback to this assessment

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ASSESSMENT CONTENT

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1 **MULTIPLE CHOICE** CORRECT 1 / 1

Consider the following conceptual sequence in an OODBMS:

1. Create object
2. Modify object
3. Program terminates

Which condition is essential for the object to remain available after the program terminates?

- A. **Correct:** The object must be made persistent in the database. — Correct answer
- B. The object must be assigned an OID by the programming language.
- C. The object must be copied into another transient object.
- D. The object must have at least one primitive attribute.

2 **MULTIPLE CHOICE** CORRECT 1 / 1

Consider the following XML snippet:

```
<product>
   <title>Wireless Noise-Cancelling Headphones</title>
```

```
    <price>1499.00</price>
</product>
```

The `<product>` element contains multiple nested child elements. Which XSD definition mechanism is required to define the structure of the product element?

- A. `xs:group`, because named model groups replace type definitions whenever elements are nested.
- B. `xs:simpleType`, because the individual nested child elements ultimately contain atomic text and numeric values.
- C. **Correct:** `xs:complexType`, because the element contains child elements rather than purely text-only content. — Correct answer
- D. `xs:restriction`, because composite elements are defined by restricting a base xs:anyType.

3 MULTIPLE CHOICE CORRECT 1 / 1

An OODBMS stores two objects with identical attribute values:

Object A:

```
studentNo = 20261234
name = "Thabo"
programme = "BSc Computer Science"
```

Object B:

```
studentNo = 20261234
name = "Thabo"
programme = "BSc Computer Science"
```

The two objects were independently created and have different object identifiers (OIDs). Which one of the following statements is most accurate?

- A. The objects are automatically merged because OODBMSs enforce value equality.
- B. The objects cannot coexist because their attribute values violate object identity constraints.
- C. The objects must represent the same database object because all their attribute values are identical.

D. **Correct:** The objects are distinct database objects because object identity is independent of their attribute values. Correct answer

4 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

Given the following query:

```
CREATE TYPE Address AS (
   street TEXT,
   city TEXT,
   postal_code TEXT
);

CREATE TABLE Customer (
   customer_id INTEGER,
   address Address
);
```

Which expression correctly accesses the `city` field of the composite value stored in address?

A. `Customer.address.city`

B. **Incorrect:** `Customer(address).city`

C. `(Customer.address).city` Correct answer

D. `address.city`

5 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

Consider the following JPA code:

```
EntityManager em = ...;
Student s = new Student();
s.setName("Mpho");
em.persist(s);
```

The database developer wants to perform a database modification using JPA. Which object is specifically associated with managing the transaction for the modification?

A. `TypedQuery`

B. **Correct:** `EntityTransaction` Correct answer

C. `PersistenceUnit`

D. `EntityManagerFactory`

6 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

Which one of the following statements best distinguishes XQuery from a simple XPath expression?

A. XQuery replaces XML Schema because it defines the structure of an XML document.

B. **Correct:** XQuery supports expressions such as FLWOR and can construct query results from XML data. Correct answer

C. XQuery is used exclusively to validate XML documents against XSD.

D. XQuery can only retrieve XML attributes, whereas XPath retrieves elements.

7 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

Consider the following function definition:

```
CREATE FUNCTION increase_price(NUMERIC)
RETURNS NUMERIC AS
$$
   SELECT $1 * 1.10;
$$ LANGUAGE SQL;
```

What does `$1` represent?

A. **Correct:** A positional parameter representing the first input argument passed to the function. Correct answer

B. An internal session variable that captures the return value of the previous statement execution.

C. An ordinal reference referring to the first column of the SELECT projection (similar to ORDER BY 1).

D. A parameter placeholder used exclusively for dynamically prepared queries via EXECUTE.

8 **MULTIPLE CHOICE** CORRECT 1 / 1

A software developer changes the following SQL:

```
SELECT *
FROM student
WHERE average > 65;
```

to JPQL.
Which conceptual transformation is required?

A. `student` must be replaced by the physical table name and average by the column name.

B. JPQL cannot express a condition on an entity attribute.

C. JPQL requires all database queries to use SELECT *.

D. **Correct:** The query must operate on the entity and its Java attribute rather than the relational table and column. Correct answer

9 **MULTIPLE CHOICE** CORRECT 1 / 1

Given the following XML structure:

```
<university>
   <student id="S1">
       <name>Katlego</name>
       <mark>72</mark>
   </student>

   <student id="S2">
       <name>Ndivhuwo</name>
       <mark>61</mark>
```

```
   </student>
</university>
```

Which XPath expression selects the names of all students whose mark is greater than 70?

A. `/university/student/name[mark > 70]`

B. `/university/student/mark[>70]/name`

C. **Correct:** `/university/student[mark > 70]/name` Correct answer

D. `/university/student[name/mark > 70]`

10 MULTIPLE CHOICE CORRECT 1 / 1

Consider a function whose logic is simply:

```
SELECT CAST(((($1 * 1000000) / $2) AS INT);
```

Which implementation is most directly appropriate?

A. **Correct:** A SQL function containing the SQL expression. Correct answer

B. A trigger function because the calculation involves parameters.

C. A PL/pgSQL function requiring DECLARE, BEGIN, and END.

D. A stored procedure because functions cannot return calculated values.

11 MULTIPLE CHOICE CORRECT 1 / 1

A university wants to restrict the value of `employment_status` to exactly:

```
FULL_TIME
PART_TIME
CONTRACT
```

No other values should be valid.

Which PostgreSQL construct most directly models this requirement?

A. **Correct:** Enumerated type — Correct answer

B. Array type

C. Geometric type

D. Composite type

12 **MULTIPLE CHOICE** CORRECT 1 / 1

Which one of the following statements is most accurate about JPQL concepts?

A. `EntityManagerFactory` is used to execute JPQL queries directly, while `EntityManager` is only used for transactions.

B. **Correct:** `EntityManager` represents the persistence context through which database operations are performed for the application. — Correct answer

C. `EntityManager` permanently stores all Java objects created by the application.

D. `EntityManagerFactory` represents an individual database transaction, while `EntityManager` represents the database schema.

13 **MULTIPLE CHOICE** CORRECT 1 / 1

Consider the following PostgreSQL query:

```
CREATE TABLE Researcher (
   name TEXT,
   publications TEXT[]
);
```

Which statement is most accurate?

A. **Correct:** `publications` stores multiple values of the array's element type. — Correct answer

B. PostgreSQL treats `publications` as a composite object containing named fields.

C. `publications` can store only one text value.

D. Every element must have a different data type.

14 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

Given the following query:

```
CREATE TABLE Staff (
   name TEXT,
   schedule TEXT[][]
);
```

A row contains:

```
{{"breakfast","meeting"},
 {"lunch","consulting"}}
```

Which expression is most appropriate for determining the dimensions of the array value?

A. array_size(schedule)

B. array_length(schedule)

C. array_dims(schedule) Correct answer

D. **Incorrect:** array_dimensions(schedule)

15 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

An OODBMS stores a `BankAccount` object containing:

```
accountNumber
balance
deposit()
withdraw()
```

A relational representation would normally store the data in columns and implement the operations separately in application code.

Which characteristic of the object model is most directly illustrated by the `BankAccount` example?

A. **Incorrect:** Object identity and independence

B. Referential integrity

C. Data independence

D. Encapsulation of state and behaviour — Correct answer

16 **MULTIPLE CHOICE** CORRECT 1 / 1

A database developer needs to implement the following logic:

```
IF salary > 50000
   calculate a 15% bonus
ELSE
   calculate a 10% bonus
END IF
RETURN the result
```

Which PostgreSQL feature is most appropriate?

A. **Correct:** A PL/pgSQL function, because procedural control structures like IF...THEN...ELSE require a procedural language handler. — Correct answer

B. A PostgreSQL stored procedure (CREATE PROCEDURE), because user-defined functions cannot return scalar values.

C. A standard SQL function, because plain SQL functions natively support procedural IF...ELSE branching blocks.

D. A trigger function, because conditional branching on table attributes can only execute within an event trigger.

17 **MULTIPLE CHOICE** CORRECT 1 / 1

Consider the following XML structure:

```
<student id="S101">
   <name>
      <first>Marlize</first>
```

```
        <last>Kruger</last>
    </name>
    <programme>COS</programme>
</student>
```

Which one of the following statements is correct?

- A. `id` is an element while first is an attribute.
- B. **Correct:** `student` is an element and `id` is an attribute of that element. — Correct answer
- C. `student` and `name` are attributes of the XML document.
- D. `name` and `programme` must be stored as XML attributes because they contain textual data.

18 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

A PostgreSQL database uses a BEFORE trigger to automatically adjust an employee's salary when a new employee record is inserted. The trigger performs the adjustment before the row is stored in the table.

Why is the row returned by a BEFORE trigger important to the triggering operation?

- A. It determines the row values that will be passed on to the triggering operation. — Correct answer
- B. It causes PostgreSQL to execute the trigger repeatedly until no changes are made.
- C. **Incorrect:** It determines whether the entire database transaction should be committed.
- D. It creates a second row independently of the row that caused the trigger to execute.

19 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

Suppose you have the following class definition:

```
class Project {
```

```
    Collection<Employee> teamMembers;
}
```

Which condition is specifically suited in JPQL to determining whether a particular employee object (`e`) occurs in a project's (`p`) collection?

- A. e IN p.teamMembers = TRUE
- B. e MEMBER OF p.teamMembers — Correct answer
- C. p.teamMembers = e
- D. **Incorrect:** p.teamMembers CONTAINS e

20 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

A database developer wants to automatically adjust an incoming salary value (e.g., `salary = salary + 500`) during an INSERT or UPDATE before the data is committed to disk.
Which trigger configuration is most appropriate?

- A. An INSTEAD OF INSERT OR UPDATE trigger, because overriding incoming column data requires replacing the default table modification mechanism.
- B. **Correct:** A BEFORE INSERT OR UPDATE trigger, because it allows direct modification of the NEW record in memory before the row is written to the table. — Correct answer
- C. An AFTER INSERT OR UPDATE trigger, because relational integrity constraints and column data types can only be validated against rows that physically exist in storage.
- D. An AFTER INSERT OR UPDATE trigger that issues a separate UPDATE statement on the table to overwrite the stored value.

ASSESSMENT CONTENT

21 **MULTIPLE CHOICE** CORRECT 3 / 3

A retail system automatically calculates the total value of an order line before the record is stored. The trigger function performs the following calculation:

```
NEW.total := NEW.quantity * NEW.unit_price;
```

The trigger is configured to execute when an order line is inserted or updated.

Select ALL statements that are correct.

☐	A. The calculation is performed only when an order line is deleted.		-33.3%
☑	B. **Correct:** The calculated value replaces the value that would otherwise have been stored in the total field.	Correct answer	33.4%
☐	C. The expression NEW.quantity * NEW.unit_price calculates the total using values from two different database tables.		-33.3%
☑	D. **Correct:** If the quantity is changed during an update, the calculation can use the new quantity supplied as part of that update.	Correct answer	33.3%
☑	E. **Correct:** If quantity = 4 and unit_price = 75, the value assigned to total is 300.	Correct answer	33.3%

22 **MULTIPLE CHOICE** CORRECT 3 / 3

Consider the following table definition:

```
CREATE TABLE WeatherStation (
   station_code TEXT,
   daily_temperatures NUMERIC[]
);
```

and the following PL/pgSQL code fragment:

```
FOREACH temp_reading IN ARRAY daily_temperatures
LOOP
   total := total + temp_reading;
END LOOP;
```

Select ALL statements that are correct.

✔	A.	**Correct:** `daily_temperatures` can store multiple numeric values within a single table column.	Correct answer	**33.3%**
✔	B.	**Correct:** The loop variable `temp_reading` captures one array element during each iteration.	Correct answer	**33.4%**
	C.	The target loop variable `temp_reading` must be declared as an array type (NUMERIC[]) rather than a scalar type to accept values from the loop.		**-33.3%**
	D.	`daily_temperatures[0]` is guaranteed to be the first element because PostgreSQL arrays always begin at index 0.		**-33.3%**
✔	E.	**Correct:** `FOREACH ... IN ARRAY` is a valid PL/pgSQL control structure used to iterate sequentially over the elements of an array.	Correct answer	**33.3%**

23 **MULTIPLE CHOICE** **CORRECT** **3 / 3**

Consider the following JPA entity definitions:

```
@Entity
class Patient {
   String fullName;
   double riskScore;

   @ManyToOne
   Ward ward;
}

@Entity
class Ward {
   String name;
}
```

A hospital administrator at Steve Biko Hospital wants to retrieve all patients with a risk score of at least 8.5 who are admitted to either the 'Intensive Care' or 'High Dependency' ward using the following JPQL query:

```
SELECT p
FROM Patient p
WHERE p.riskScore >= 8.5
AND (
   p.ward.name = 'Intensive Care'
   OR p.ward.name = 'High Dependency'
```

)

Select ALL statements that are correct regarding this query:

- [ ] **A.** The query returns only the scalar `fullName` attribute of each matching patient. **-33.3%**
- [x] **B.** **Correct:** The parentheses ensure that the two OR ward conditions are evaluated together before combining with the score predicate. Correct answer **33.4%**
- [x] **C.** **Correct:** `p` is a JPQL identification variable representing individual instances of the Patient entity. Correct answer **33.3%**
- [x] **D.** **Correct:** `p.ward.name` illustrates a path expression that navigates across an entity relationship. Correct answer **33.3%**
- [ ] **E.** Replacing `SELECT p` with `SELECT *` produces equivalent, valid JPQL syntax. **-33.3%**

24 **MULTIPLE CHOICE** **CORRECT** **3 / 3**

Consider the following XML document:

```
<airline>
   <flight code="SA204">
       <destination>Frankfurt</destination>
       <seats_available>45</seats_available>
   </flight>
   <flight code="SA309">
       <destination>Nairobi</destination>
       <seats_available>12</seats_available>
   </flight>
</airline>
```

Select ALL statements that are correct regarding the XML document and XML technologies.

- [ ] **A.** The destination and seats_available values are XML attributes of their respective flight elements. **-33.3%**

☑ B.	**Correct:** The flight elements are child elements of the airline element.	Correct answer	**33.3%**
☑ C.	**Correct:** XML Schema can be used to specify the structure and data types permitted in an XML document.	Correct answer	**33.4%**
☐ D.	XML requires every element to have at least one child element.		**-33.3%**
☑ E.	**Correct:** The value SA204 is an attribute value associated with the flight element.	Correct answer	**33.3%**

25 **MULTIPLE CHOICE** **CORRECT** **3 / 3**

Consider the following statements about an OODBMS and the ObjectDB/JPA approach.
Select ALL statements that are correct.

☑ A.	**Correct:** Operations that modify database content using ObjectDB/JPA should be performed within an active EntityTransaction.	Correct answer	**33.4%**
☑ B.	**Correct:** In JPA, an EntityManager is used to perform database operations on persistent objects.	Correct answer	**33.3%**
☐ C.	JPQL queries are written primarily in terms of physical database tables and columns.		**-33.3%**
☑ D.	**Correct:** An object can have an identity that is independent of the values of its attributes.	Correct answer	**33.3%**
☐ E.	Making an object persistent necessarily means that all objects referenced by it are automatically converted into relational foreign-key records.		**-33.3%**

ASSESSMENT CONTENT

26 **FILL IN THE BLANK** **PARTIALLY CORRECT** **2 / 3**

A university uses a PostgreSQL function to determine whether a laboratory sample queue has been completely processed. When its pending, the status must be set to 'PENDING'. And then the status is completed it must be set to 'COMPLETED'. Complete the function such that it continues processing while samples remain.

```
CREATE FUNCTION process_queue(samples INTEGER)
RETURNS TEXT AS $$
DECLARE
   remaining INTEGER := samples;
   status TEXT := 'PENDING';
BEGIN
   WHILE ✓ remaining > 0 LOOP
       remaining := remaining - 1;
   END LOOP;
   status := ✗ "COMPLETE";
   RETURN ✓ status;
END;
$$ LANGUAGE plpgsql;
```

Partial Credit

Acceptable answers for **BLANK-1** are **remaining >0**, **remaining > 0**, **remaining> 0** and **remaining>0**
Responses must match exactly

Acceptable answers for **BLANK-2** are **'COMPLETED'**, **'COMPLETE'** and **'PROCESSED'**
Responses must match exactly

Acceptable answer for **BLANK-3** is **status**
Responses must match exactly

27 **FILL IN THE BLANK** **INCORRECT** **0 / 4**

A university maintains equipment calibration records.

```
CREATE TABLE Equipment (
   equipment_id INTEGER,
   calibration_score NUMERIC,
   status TEXT
);
```

The university requires that equipment with a calibration score of 80 or higher is automatically assigned the status CERTIFIED. Equipment with a lower score must be assigned the status REVIEW.
The trigger function below is incomplete. Complete each blank.

```
CREATE FUNCTION determine_status()
RETURNS TRIGGER AS $$
```

```
BEGIN
    IF ✗calibration_score THEN
        ✗status := 'CERTIFIED';

    ELSE
        ✗status := 'REVIEW';
    END IF;
    RETURN ✗status;

END;
$$ LANGUAGE plpgsql;
```

Incorrect

Acceptable answers for **BLANK-1** are **NEW.calibration_score >= 80**, **NEW.calibration_score > 79** and **NEW.calibration_score>=80**
Responses must match exactly

Acceptable answer for **BLANK-2** is **NEW.status**
Responses must match exactly

Acceptable answer for **BLANK-3** is **NEW.status**
Responses must match exactly

Acceptable answer for **BLANK-4** is **NEW**
Responses must match exactly

28 **FILL IN THE BLANK** CORRECT 3 / 3

A university's postgraduate research system stores `Researcher` objects in ObjectDB.
Assume the following objects have already been persisted:

Researcher	Research Area	Years
R001	Cybersecurity	5
R002	Databases	3
R003	Cybersecurity	7
R004	Digital Forensics	4

(a) The following JPQL query is executed:

```
Query q = em.createQuery(
    "SELECT COUNT(r) FROM Researcher r " +
    "WHERE r.years >= 5"
);
```

What value is returned by (if nothing state “nothing”)? (1 Mark):

```
q.getSingleResult()
```

Answer: ✓ 2

(b) The following query is then executed:

```
Query q2 = em.createQuery(
   "SELECT AVG(r.years) FROM Researcher r " +
   "WHERE r.researchArea = 'Cybersecurity'"
);
```

What value is returned (if nothing state “nothing”)? (1 Mark):

Answer: ✓ 6

(c) Complete the following JPQL statement so that it returns the research area of every researcher with at least 4 years of experience.

```
"SELECT r.researchArea " +
"FROM Researcher r " +
"WHERE r.✓ years >= 4 "
```

Correct

Acceptable answer for **BLANK-1** is **2**
Responses must match exactly

Acceptable answers for **BLANK-2** are **nothing** and **6**
Responses must match exactly

Acceptable answers for **BLANK-3** are **years>=4**, **years >= 4**, **years >=4**, **years>= 4** and **years > = 4**
Responses must match exactly

29 **FILL IN THE BLANK** **PARTIALLY CORRECT** 1 / 3

A museum database stores artefact locations using the following composite type:

```
CREATE TYPE Location AS (
   building TEXT,
   floor_no INTEGER,
   room_no INTEGER
);

CREATE TABLE Artefact (
   artefact_id INTEGER,
   description TEXT,
   location Location
);
```

The following rows exist:

artefact_id	description	location
501	Bronze Mask	(Heritage Centre, 2, 17)

502	Iron Spear	(Heritage Centre, 2, 21)
503	Gold Coin	(Archaeology Wing, 1, 17)

(a) The museum wants to retrieve the building in which artefact 501 is located. Complete the expression (1 Mark):

```
SELECT ✓ (location).building
FROM Artefact
WHERE artefact_id = 501;
```

(b) The museum wants to retrieve all artefacts located on **floor 2**. Complete the condition (1 Mark):

```
SELECT artefact_id, description
FROM Artefact
WHERE ✗ (location).floor = 2;
```

(c) The room number of artefact 503 must be changed from 17 to 25. Complete the composite attribute name in the following statement (1 Mark):

```
UPDATE Artefact
✗ (location).room_no ✗ = 25
WHERE artefact_id = 503;
```

Partial Credit

Acceptable answers for **BLANK-1** are **(location).building** and **(Artefact.location).building**
Responses must match exactly

Acceptable answers for **BLANK-2** are **(location).floor_no = 2** and **(Artefact.location).floor_no = 2**
Responses must match exactly

Acceptable answer for **BLANK-3** is **SET (location).room_no**
Responses must match exactly

30 **FILL IN THE BLANK** **PARTIALLY CORRECT** 2 / 3

A music streaming service stores playlists as XML:

```
<library>
 <playlist name="SA Hits 2026">
     <song>
         <title>Too Cool</title>
         <artist>Uncle Waffles</artist>
         <duration>368</duration>
     </song>
     <song>
         <title>Izospana</title>
         <artist>Sam Deep</artist>
         <duration>443</duration>
     </song>
     <song>
         <title>Khanya Njalo</title>
         <artist>Kabza De Small</artist>
         <duration>509</duration>
     </song>
     <song>
         <title>Ngiyak'cela</title>
         <artist>Shaunmusiq</artist>
         <duration>531</duration>
     </song>
 </playlist>
</library>
```

(a) Write an XPath expression that returns all song titles (1 Mark).
Answer: ✓ **//playlist/song/title**

(b) Write an XPath expression that returns the titles of songs longer than 400 seconds (1 Mark).
Answer: ✓ **//playlist/song[duration > 400]/title**

(c) Write an XPath expression to retrieve the artist of the song titled "Too Cool":
Answer: ✗ **//playlist/song/title[artist = "Too Cool"]**

Partial Credit

Acceptable answers for **BLANK-1** are **/library/playlist/song/title**, **//playlist/song/title**, **//playlist/song/title/text()**, **/library/playlist/song/title/text()**, **//library/playlist/song/title/text()** and **//library/playlist/song/title**
Responses must match exactly

Acceptable answers for **BLANK-2** are **/library/playlist/song[duration > 400]/title**, **//playlist/song[duration > 400]/title**, **//playlist/song[duration > 400]/title/text()**, **/library/playlist/song[duration > 400]/title/text()**, **//library/playlist/song[duration > 400]/title/text()**, **//library/playlist/song[duration > 400]/title**, **//library/playlist/song[duration>400]/title**, **//playlist/song[duration>400]/title**,

/library/playlist/song[duration>400]/title/text() and **/library/playlist/song[duration>400]/title**
Responses must match exactly

Acceptable answers for **BLANK-3** are **/library/playlist/song[title = 'Too Cool']/artist**, **//playlist/song[title = 'Too Cool']/artist/text()**, **//playlist/song[title = 'Too Cool']/artist**, **/library/playlist/song[title = 'Too Cool']/artist/text()**, **/library/playlist/song[title="Too Cool"]/artist**, **//library/playlist/song[title = "Too Cool"]/artist/text()**, **//library/playlist/song[title = "Too Cool"]/artist**, **/library/playlist/song[title = "Too Cool"]/artist**, **//library/playlist/song[title = 'Too Cool']/artist**, **//library/playlist/song[title="Too Cool"]/artist**, **//playlist/song[title="Too Cool"]/artist** and **/library/playlist/song[title = "Too Cool"]/artist/text()**
Responses must match exactly

31 **FILL IN THE BLANK** CORRECT 3 / 3

Consider the following XML Schema:

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="PersonType">
      <xs:sequence>
          <xs:element name="firstName" type="xs:string"/>
          <xs:element name="age" type="xs:integer"/>
      </xs:sequence>
  </xs:complexType>

  <xs:element name="person" type="PersonType"
              minOccurs="1" maxOccurs="3"/>

</xs:schema>
```

(a) The `PersonType` definition can be referenced by other elements in the schema. What type of XSD definition is `PersonType`(1 Mark)?
Answer: ✓ **global**

(b) The firstName element contains text and does not contain child elements. What type of XML Schema type is being used for firstName (1 Mark)?
Answer: ✓ **simple**

(c) A valid XML document conforming to this schema contains three `<person>` elements. What is the total number of `<age>` elements that must occur in the document (1 Mark)?
Answer: ✓ **3**

Correct

Acceptable answers for **BLANK-1** are **Global type**, **Global** and **GlobalType**
Responses must match exactly

Acceptable answers for **BLANK-2** are **Simple type**, **Simple** and **simpletype**
Responses must match exactly

Acceptable answers for **BLANK-3** are **3** and **Three**
Responses must match exactly

32 **FILL IN THE BLANK** **PARTIALLY CORRECT** 2 / 3

A developer creates a Device object for a laboratory management application.

```
Device d = new Device();
d.setSerialNo("LAB-4821");
d.setStatus("AVAILABLE");
em.getTransaction().begin();
em.persist(d);
d.setStatus("ALLOCATED");
em.getTransaction().commit();
```

Assume no other code modifies the object.
(a) What value will be stored for status after the transaction commits? (1 Mark)
Answer: ✓ **ALLOCATED**

(b) Complete the following statement:
em.✗ **close** ();
to permanently complete the transaction (1 Mark).

(c) Suppose the developer inserts `em.clear();` immediately after `em.persist(d);` and before `d.setStatus("ALLOCATED");`.
What value for status will now be saved in the database when the transaction commits? (1 Mark)
Answer: ✓ **AVAILABLE**

Partial Credit

Acceptable answers for **BLANK-1** are **ALLOCATED** and **"ALLOCATED"**
Responses must match exactly

Acceptable answer for **BLANK-2** is **getTransaction().commit**
Responses must match exactly

Acceptable answers for **BLANK-3** are **AVAILABLE** and **"AVAILABLE"**
Responses must match exactly

33 **FILL IN THE BLANK** PARTIALLY CORRECT 2 / 3

A car rental company uses ObjectDB with JPA to store persistent Vehicle objects. The following entity class is used:

```
 @Entity
public class Vehicle {
   @Id
   private String registrationNo;
   private String category;
   private double dailyRate;
   private boolean available;

   // constructors and methods omitted
}
```

The application must establish access to the ObjectDB database before it can store objects.

Complete the following statements:
(a) Complete the statement used to create an `EntityManagerFactory` for the persistence unit named `vehicle.odb (1 Mark)`:

`EntityManagerFactory emf =(` ✗ **jpaConstructor(/** `"vehicle.odb");`

(b) Complete the statement used to obtain an EntityManager from the EntityManagerFactory:
`EntityManager em = emf.(` ✓ **createEntityManager(** `); (1 Mark)`

(c) A transaction is required before making changes to the database. Complete the statement used to obtain the transaction associated with the EntityManager:
`EntityTransaction tx = em.` ✓ **getTransaction** `();`

Partial Credit

Acceptable answers for **BLANK-1** are **Persistence.createEntityManagerFactory** and **Persistence.createEntityManagerFactory(**
Responses must match exactly

Acceptable answers for **BLANK-2** are **createEntityManager** and **createEntityManager(**
Responses must match exactly

Acceptable answer for **BLANK-3** is **getTransaction**
Responses must match exactly

ASSESSMENT CONTENT

34 **FILL IN THE BLANK** **INCORRECT** **0 / 1**

A Kruger National Park wildlife conservation application stores two Elephant objects.

Elephant A

```
OID: 7821
name: "Tembo"
age: 34
```

Elephant B

```
OID: 9145
name: "Tembo"
age: 34
```

Both objects have identical attribute values.

What makes the two objects **distinct objects** in the OODBMS?

Answer: ✗ **Their respective OID**

Incorrect

Acceptable answers for **BLANK-1** are **OID** and **Object identifier**

Responses must match exactly

35 **FILL IN THE BLANK** **CORRECT** **1 / 1**

A PL/pgSQL function needs to create and initialise variables before the executable statements in the function body are processed.

What is the name of the section used to define these variables? (1 Mark)

Answer: ✓ **DECLARE**

Correct

Acceptable answer for **BLANK-1** is **Declare**

Responses must match exactly

36 **FILL IN THE BLANK** **CORRECT** **1 / 1**

An organisation uses XML documents from different systems, and two systems use the same element name for different purposes. The XML documents therefore need a mechanism to ensure that element names from different XML vocabularies do not conflict.

What XML concept is used to prevent these naming conflicts?

Answer: ✓ **namespace**

Correct

Acceptable answers for **BLANK-1** are **Namespace** and **Namespaces**
Responses must match exactly

37 **FILL IN THE BLANK** INCORRECT 0 / 1

A car-rental application uses ObjectDB/JPA. The following Vehicle entity is stored in the database:

```
 @Entity
public class Vehicle {
   @Id
   private Long vehicleId;
   private String registrationNo;
   private String category;
   private double dailyRate;

   // constructors and methods omitted
}
```

The application needs to retrieve the Vehicle object whose primary key is 4821L.

Complete the following statement:
`Vehicle v = em.` ✗ **findObject** `(Vehicle.class, 4821L);`

Incorrect

Acceptable answer for **BLANK-1** is **find**
Responses must match exactly

38 **FILL IN THE BLANK** INCORRECT 0 / 1

The ✗ **XSD** XML document type acts like a digital transfer file primarily built for machines to read and exchange information smoothly (1 Mark).

Incorrect

Acceptable answers for **BLANK-1** are **Data-centric** and **Data centric**
Responses must match exactly

39 **FILL IN THE BLANK** CORRECT 1 / 1

A wildlife conservation system stores persistent Animal objects. The application needs to retrieve all Animal objects using JPQL.
Complete the following statement:

```
TypedQuery<Animal> query = em.createQuery("SELECT a FROM Animal a",
```
✓ **Animal.class**) ;

Correct

Acceptable answer for **BLANK-1** is **Animal.class**
Responses must match exactly

40 **FILL IN THE BLANK** INCORRECT 0 / 1

An OODBMS administrator wants object identities to remain valid even when objects are moved to different physical locations on disk. The identifier should identify the object independently of where the object is physically stored. This requirement is satisfied by a ✗ **persistent** OID.

Incorrect

Acceptable answer for **BLANK-1** is **logical**
Responses must match exactly

41 **FILL IN THE BLANK** INCORRECT 0 / 1

A university database uses relations to represent students, modules and registrations. However, the relational model itself does not explicitly distinguish between the different types of relationships represented by these structures. The limitation illustrated by this situation is known as ✗ **referntial integrity**.

Incorrect

Acceptable answer for **BLANK-1** is **semantic overloading**
Responses must match exactly

42 **FILL IN THE BLANK** CORRECT 1 / 1

A banking system processes a withdrawal request. A withdrawal must not be allowed when the requested amount is greater than the customer's available balance.
Complete the missing PL/pgSQL statement:

```
DO $$
```

```
DECLARE
   balance NUMERIC := 2500;
   withdrawal NUMERIC := 3000;
BEGIN
   IF withdrawal > balance THEN
      ✓ RAISE EXCEPTION 'Insufficient funds for withdrawal';
   END IF;
END;
$$ LANGUAGE plpgsql;
```

Correct

Acceptable answer for **BLANK-1** is **RAISE EXCEPTION**
Responses must match exactly

---

43 **FILL IN THE BLANK** CORRECT 1 / 1

A Formula 1 timing system stores a driver's latest lap position as a composite value:

```
CREATE TYPE LapPosition AS (
   lap INTEGER,
   sector INTEGER,
   position INTEGER
);
```

The table contains:

```
CREATE TABLE Driver (
   driver_name TEXT,
   latest LapPosition
);
```

Complete the expression:

```
SELECT ✓ (latest).position
FROM Driver;
```

to retrieve the driver's current track position.

Correct

Acceptable answers for **BLANK-1** are **(latest).position** and **(Driver.latest).position**
Responses must match exactly

ASSESSMENT CONTENT

Study the following scenario and answer the following questions:

Wamanzi pvt Ltd is a marine research company based in Durban. It operates a fleet of autonomous underwater vehicles (AUVs) used to explore deep-sea environments. Each AUV has:

- `auvId`
- `model`
- `mission`
- `depthCapability`
- `active`

During a mission, an AUV collects many **SensorReading** objects. Each reading contains:

- `temperature`
- `pressure`
- `depth`
- `timestamp`

The application developers decide to use ObjectDB/JPA because an AUV and its sensor readings form an object graph that must be persisted.

The operational database is PostgreSQL. Each AUV stores its energy consumption for the last six mission stages in an array. The system must calculate the average energy consumption whenever a record is inserted or updated.

The AUVs also transmit mission summaries as XML.

A mission summary has the following structure:

```
<mission>
   <missionCode>M884</missionCode>
   <auvId>A17</auvId>
   <status>COMPLETED</status>
   <readings>
       <reading>
           <depth>4200</depth>
           <temperature>2.4</temperature>
       </reading>
       ...
   </readings>
</mission>
```

Mission managers need to retrieve particular information from these XML documents.

Source: Makura S.M (2026)

(a) OODBMS / JPA - 7 marks

Write a JPA entity class for AUV.

Your solution must:

1. Declare the class as an entity.
2. Define auvId as the entity identifier.

3. Define the remaining attributes appropriately (no need for constructor, getters and setters).
4. Represent the relationship between an AUV and its multiple `SensorReading` objects.
5. Write a JPQL query that retrieves the IDs of all active AUVs whose depth capability exceeds 4000 metres.

(b) ORDBMS / PostgreSQL - 7 marks

PostgreSQL contains:

```
CREATE TABLE AUVEnergy (
   auv_id INTEGER PRIMARY KEY,
   stage_energy NUMERIC[],
   average_energy NUMERIC
);
```

The developers want `average_energy` to be calculated automatically from `stage_energy`.

Write the PostgreSQL implementation that:

1. Creates a PL/pgSQL function returning TRIGGER.
2. Iterates through the values in `NEW.stage_energy`.
3. Calculates the average energy consumption.
4. Stores the calculated value in `NEW.average_energy` and returns `NEW`.
5. Creates an appropriate `BEFORE INSERT OR UPDATE` row-level trigger.

You may assume that `stage_energy` contains at least one value.

(c) XML / XPath / XQuery - 6 marks

The mission-control application needs to analyse the XML mission summaries.

Using the XML structure given above:

(i) Write an XPath expression that retrieves the depth of every reading greater than 3000 metres. (2)

(ii) Write an XSD fragment defining the `missionCode` element as a required textual element and the status element as a textual element. Your answer must use the appropriate XML Schema constructs. (4)

TOTAL: 20 MARKS

Answer

a)

@Entity
Public Class AUV{

  @Id
  private String auvId;
  private String model;
  private String model;
  private int depthCapability;
  private Boolean active;

```
  @oneToMany
  SensorReading sensorReading;

(5: (I assume you just want the query itself not the function since there is no real function
name specified) -
"SELECT a.auvId FROM AUV a WHERE a.depthCapability > 4000"
}

@Entity
Public Class SensorReading{

  private Double temperature;
  private Double pressure;
  private int depth;
  private String timestamp;

}
-------------------------------------------------------------------------------------------------------------------------
-------------------
b)

CREATE FUNCTION average_energy() RETURN TRIGGER AS
$$
DECLARE
val NUMERIC
count int := 0
sum NUMERIC := 0

BEGIN

FOR EACH val IN ARRAY NEW.state_energy
  LOOP
```